Autoantibodies Encountered in Patients with Autoimmune Connective Tissue Diseases

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Key features

- Autoantibodies can play an important role in the diagnosis, management, and prognosis of autoimmune connective tissue diseases (AI-CTDs)
- The classic anti-nuclear antibody (ANA) assay remains the entrée into the world of AI-CTD serology, and understanding its limitations is critical for clinical decision-making
- While many autoantibodies are not disease-specific, certain autoantibodies have significant disease specificity and therefore are of diagnostic value
- Anti-dsDNA and anti-Smith (Sm) antibodies have high specificity for systemic lupus erythematosus (SLE)
- Anti-Mi-2, -TIF1-γ, -MDA5, -NXP-2, and -SAE antibodies are associated with different forms of dermatomyositis
- Anti-Jo-1 antibodies can be detected in antisynthetase syndrome
- Anti-topoisomerase-1 (Scl-70), -RNA polymerase III, and -centromere antibodies are associated with different clinical forms of systemic sclerosis
- Cytoplasmic anti-neutrophil cytoplasmic antibodies (c-ANCA) are often detected in granulomatosis with polyangiitis

INTRODUCTION

Autoimmune connective tissue diseases (AI-CTDs) represent polygenic clinical disorders with heterogeneous and overlapping clinical features. AI-CTDs are characterized by an immune dysregulation that includes the production of antibodies to self-antigens, termed autoantibodies (aAb). When considering the diagnosis of an AI-CTD, the most commonly requested laboratory test is the measurement of anti-nuclear antibodies (ANA). ANA are defined as aAb that target primarily nuclear components, including DNA or small nuclear ribonucleoproteins (snRNP). Elevated ANA titers are seen in multiple AI-CTDs and they reflect inflammatory processes within tissue but rarely have pathogenic potential.

Clinical laboratory testing for ANA is essential in the diagnosis and management of AI-CTD patients[1]. Each AI-CTD has aAb that are quite specific to the particular disorder as well as aAb that are shared with other diseases. Although AI-CTDs are usually distinct clinical entities, some patients may have overlapping features of two or more AI-CTDs and, as a result, an overlap in their pattern of aAb production (Fig. 40.1). Familiarity with the actual serologic tests and the disease associations of the various aAb allows one to maximize the clinical utility of aAb testing.

This chapter discusses aAb associated with AI-CTDs that have cutaneous manifestations in addition to historical and technical aspects of ANA assays. Tables 40.1–40.4 summarize the molecular identity of the various autoantigens and their clinical associations. The prevalence of the aAb in these tables represents the estimated median values reported from different cross-sectional studies using conventional clinical laboratory testing procedures. However, the results of such studies can vary widely. In addition, there can be complex relationships between the presence and relative amounts of aAb, disease diagnosis, and prognosis. Because understanding the basic principles of aAb laboratory tests such as sensitivity and specificity assists in interpreting assay results, they are also reviewed. Chapters 4, 23, and 24 discuss the related subjects of circulating immune complexes and cryoglobulins while Chapters 41–45 contain an in-depth discussion of the various AI-CTDs.

HISTORICAL PERSPECTIVE

One can better appreciate the evolution of thought in this area by considering the changes in methodologies used to identify aAb that have occurred over time[2]. In 1948, the bone marrow from a patient with SLE was noted to contain polymorphonuclear neutrophils that had phagocytosed nuclear material from degenerated cells. The presence of these cells became an indicator of SLE (SLE cell phenomenon)[3] and occurred in the presence of antibodies to DNA.

Approximately 10 years later, indirect immunofluorescence techniques were used to detect ANA, with the fluorescent anti-nuclear antibody (FANA) test providing a more sensitive assay for SLE. Another decade later, the Ouchterlony double immunodiffusion (DID) technique was used to identify and define precipitating serum aAb that reacted with saline extractable nuclear antigens (ENA) such as nuclear ribonucleoprotein (nRNP), Sm, SSA/Ro, and SSB/La[4].

In 1979, the molecularly and genetically defined ANA era began as the result of work by Lerner and Steitz, who employed immunodiffusion and Western blotting to define the molecular identities of the nRNP and Ro/La families of ribonucleoprotein autoantigens. These observations generated the notion that aAb might be more efficiently and cost-effectively assayed by using purified or recombinant forms of the different autoantigens in solid-phase immunoassays. An example of the latter is the *e*nzyme-*l*inked *i*mmuno*s*orbent *a*ssay (ELISA). While less expensive and more convenient, there are some drawbacks to employing ELISA and related immunoassay techniques to detect ANA, e.g. increased sensitivity and decreased specificity for aAb such as anti-SSA/Ro and anti-dsDNA antibodies. Newer multiplex techniques (e.g. solid-phase multiplex bead assays, proteome microarrays, Luminex® xMAP® technology) are now widely used and allow the simultaneous profiling of all relevant aAb in a patient. The principles of these assays are shown in Fig. 40.2, and their limitations are discussed in a recent review[5].

THE FLUORESCENT ANTI-NUCLEAR ANTIBODY (FANA) TEST

Although there is debate about its role in the diagnosis and management of AI-CTD, the classic fluorescent ANA assay (i.e. FANA test) is still the most clinically efficient screening test for systemic autoimmune disorders[6]. The ANA assay identifies antibodies present in serum that bind to autoantigens present in the nuclei (or cytoplasm) of mammalian cells. Both the pattern, which reflects the localization of the autoantigen, and the titer are reported. The reported titer represents the last dilution at which the ANA pattern is detectable and a titer of ≥1 : 40 is considered "positive". For the FANA test, almost all clinical laboratories currently employ a human tumor cell line such as HEp-2 for the nucleated cell substrate. aAb are detected with fluorochrome-conjugated antibodies that bind human immunoglobulin (the aAb) bound to nuclei in the HEp-2 cells (see Fig. 40.2A).

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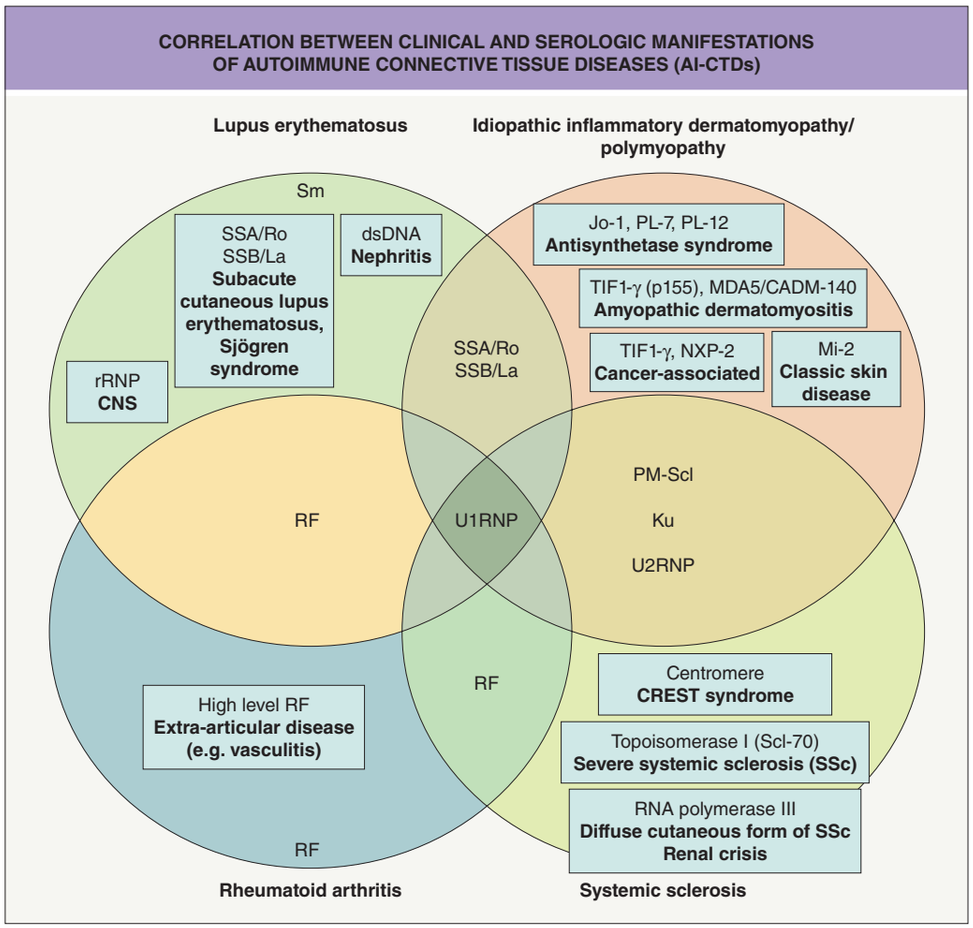

Fig. 40.1 Correlation between clinical and serologic manifestations of autoimmune connective tissue diseases (AI-CTDs). Venn diagram representation of major clinico-serologic correlations in the primary AI-CTDs that impact the skin. Some targets of autoantibodies are specific (boxed with clinical associations) while others are seen in the context of overlapping clinical features. High levels of U1RNP autoantibody with overlapping features of all four of these AI-CTDs defines the concept of mixed connective tissue disease. CADM, clinically amyopathic dermatomyositis; MDA5, melanoma differentiation-associated protein 5; p155, a 155 kDa polypeptide; RF, rheumatoid factor; RNP, ribonucleoprotein, TIF1-γ, transcriptional intermediary factor 1-gamma.

Earlier versions of the ANA assay utilized rodent cell lines that lack some of the autoantigens present in human cell nuclei (e.g. SSA/Ro). Thus, sera from SLE patients could be negative when assayed on rodent cells, especially if the predominant aAb were anti-SSA/Ro (i.e. "ANA-negative SLE"). In the past, this occurred in up to 15% of SLE patients, particularly those enriched in anti-SSA/Ro aAb-associated disorders such as subacute cutaneous LE (SCLE) and Sjögren syndrome (SjS). With the use of human HEp-2 cells, only ~1%–2% of SLE patients are ANA-negative, and "ANA-negative SLE" has almost become a historical phenomenon.

An important caveat is that the *titer* (which reflects the serial serum dilutions necessary for fluorescence to disappear) depends upon subjective interpretation by a laboratory technician. Consequently, a single serum specimen can have an ANA titer result that varies within a two-tube dilution range. In other words, a serum sample reported as having an ANA titer of 1:320 might be read as 1:160 or 1:640 upon repeat testing in the same laboratory. Although the World Health Organization tried to standardize reporting of ANA results with an international unit system (e.g. 1 IU rather than a titer of 1:160), many laboratories continue to report ANA results using a titer system. Of note, while an ANA titer of ≥1:40 is considered "positive", the 2019 European League Against Rheumatism/American College of Rheumatology (EULAR/ACR) criteria require a titer of ≥1:80 for the diagnosis of SLE (see Table 41.5).

Automated Immunoassays – Solid Phase and Bead-Based Multiplex

Automated systems have been introduced to decrease variability and increase cost-effectiveness (see Fig. 40.2B,C); however, pattern reporting remains an issue[7]. In addition, concerns exist that the results from the FANA assay may not be comparable to those of ELISA- or multiplex bead-based assays. While the FANA test has a consistently high sensitivity, this can be more variable amongst different ELISA or bead-based assays, likely due in part to the requirement of the latter to have predefined antigen combinations for ANA identification. Use of such predetermined antigen combinations is not a component of the FANA test.

Compared to the quantitative ELISA, multiplex bead-based assays have reduced labor requirements and increased throughput, but require specialized equipment. The FANA assay has the drawback of requiring significant experience for proper interpretation as well as the need for subsequent tests for antigen identification and quantitation.

"Normal" Versus "Abnormal" ANA Titers

The ANA titer may vary significantly, depending upon how the assay is performed and interpreted (see above). Major commercial laboratories typically report three ANA categories: (1) negative (titer <1:40); (2) equivocal (1:40 to 1:80); and (3) positive (>1:80). However, some laboratories will record an ANA titer of 1:40 as abnormal, presumably because accepting such low levels of ANA as being abnormal retains a high degree of *sensitivity* in detecting systemic AI-CTD. However, using such low ANA titer cut-offs can lead to positive, but clinically insignificant, ANA test results; i.e., it reduces specificity. Several studies that have compared ANA results in SLE populations with those in normal control populations have indicated that a titer of <1:160 has little clinical utility. For example, in one report based upon 15 international laboratories, the ANA positivity rate in a population of healthy individuals (ages 20–60 years) was 13% at 1:80, 5% at 1:160, and 3% at 1:320[8]. Elevated ANA titers can be observed in the elderly, relatives of those with SLE, and patients with other autoimmune disorders (e.g., autoimmune thyroid disease), as well as in healthy individuals. Potential pitfalls when interpreting the results of ANA assays are summarized in Table 40.5.

Immunofluorescence Patterns

Various patterns of nuclear immunofluorescence can be observed in the FANA test (Fig. 40.3): homogeneous (syn. diffuse), peripheral (syn. rim), speckled (syn. particulate), nucleolar, and centromeric (syn. discrete speckled). Except for the centromeric pattern, these patterns

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AUTOANTIBODIES ASSOCIATED WITH LUPUS ERYTHEMATOSUS			
Target	**Median prevalence***	**Molecular specificity**	**Clinical associations**
High specificity for SLE			
dsDNA	60%–90%	Double-stranded (native) DNA	LE nephritis and monitoring activity of nephritis
Sm	10%–30% of White population 30%–40% of Asians and African-Americans	Spliceosome RNP – RNP particles involved in splicing pre-mRNA	Renal, neurologic, and hematologic involvement
rRNP	7%–15% 40% of Asians	Ribosomal P proteins – involved in ribosome function	Neuropsychiatric LE
Low specificity for SLE			
ANA (most common IF patterns: homogeneous, peripheral)	99%		
ssDNA	70%–95%	Denatured DNA	Possible risk for SLE in DLE patients; also seen in RA, DM/PM, MCTD, SSc, SjS, morphea
Histones	30%–70%	Histones	Drug-induced SLE; also occur in SLE, RA, and SSc with pulmonary fibrosis (in conjunction with other aAb)
C1q	20%–50% Nearly 100% of patients with active lupus nephritis	C1q component of complement	LE nephritis, severe SLE, hypocomplementemic urticarial vasculitis syndrome
U1RNP	30%–50%	Spliceosome RNP	Overlapping features with other AI-CTDs; MCTD (100%)
SSA/Ro	30%–50%†	Ro60 (60 kDa protein): hYRNP – quality control function for misfolded RNA molecules Ro52 (52 kDa protein): also known as TRIM21 – interferon-inducible protein with ubiquitin modifying function	SCLE (75%–90%), neonatal LE/congenital heart block (99%), SCLE–SjS overlap, primary SjS (70%); in SjS associated with vasculitis
SSB/La	10%–20%	hYRNP	SCLE (30%–40%), SCLE–SjS overlap, primary SjS (40%); occurs in conjunction with SSA/Ro
Cardiolipin	IgM: 20%–75% IgG: 15%–30%‡	Cardiolipin – a negatively charged phospholipid	Recurrent spontaneous abortions, thrombocytopenia, and hypercoagulable state in SLE (cutaneous manifestations include livedo reticularis, leg ulcers, acral infarction/ulceration, hemorrhagic cutaneous necrosis); similar associations in primary antiphospholipid antibody syndrome (APS); clinical manifestations have strongest association with IgG class of anti-cardiolipin aAb
β2 glycoprotein I	IgM: 10%–35% IgG: 8%–25%‡	An important cofactor for cardiolipin in cardiolipin aAb assays	Relatively high risk of thrombosis in SLE and primary APS
"Non-criteria" phospholipids	Phosphatidylserine IgG: 20% of SLE patients with thrombosis Phosphatidylinositol IgG: 20% of SLE patients with thrombosis Phosphatidylcholine: rare Phosphatidylethanolamine: rare Annexin V: rare	Negatively charged phospholipids	Clinical significance remains under investigation; not currently included in SLE or APS clinical criteria
Rheumatoid factor	25%	Fc portion of IgG	Nonspecific
Ku	10%	DNA end-binding repair protein complex	Overlap with other AI-CTDs such as DM/PM, SSc
Alpha-fodrin	10%	An actin-binding protein found at the periphery of chromaffin cells that may be involved in secretion	SjS

*Based on most common assay techniques currently employed in clinical immunology laboratories.
†Refers to SSA/Ro antigen detection as a whole and not Ro52 or Ro60 subsets; see text for specific Ro52 and Ro60 associations.
‡Range depending on 99th or 90th percentile assay cut-off.

Table 40.1 Autoantibodies associated with lupus erythematosus. aAb, autoantibody; AI-CTDs, autoimmune connective tissue diseases; ANA, anti-nuclear antibodies; DLE, discoid lupus erythematosus; DM/PM, dermatomyositis/polymyositis; IF, immunofluorescence; LE, lupus erythematosus; MCTD, mixed connective tissue disease; PCNA, proliferating cell nuclear antigen; RA, rheumatoid arthritis; RNP, ribonucleoprotein; SCLE, subacute cutaneous lupus erythematosus; SjS, Sjögren syndrome; SLE, systemic lupus erythematosus; SSc, systemic sclerosis. Adapted from references 2, 16, 51–53.

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AUTOANTIBODIES ENCOUNTERED IN THE IDIOPATHIC INFLAMMATORY DERMATOMYOPATHIES			
Target	**Median prevalence***	**Molecular specificity**	**Clinical associations**
High specificity for DM/PM			
Mi-2	Classic DM: 10%–35% Juvenile DM: 4%–10%	Helicase nuclear protein	• Gottron papules/sign, shawl sign, cuticular telangiectasias, cuticular overgrowth/dystrophy • Favorable response to therapy and better prognosis
TIF1-γ (p155)	Classic DM: 10%–30% Clinically amyopathic: 20%–45% Classic DM in Japanese adults: 50% Juvenile DM: 20%–35%	p155: transcriptional intermediary factor 1 gamma (TIF1-γ) p140: TIF1-α	• Clinically amyopathic DM • In adult-onset classic DM, increased risk of malignancy • Extensive photosensitive cutaneous disease • Mucocutaneous findings include palatal erythema ("ovoid patches"), psoriasiform lesions, and hypopigmented and telangiectatic ("red on white") patches • In juvenile DM, associated with treatment-refractory disease, ulcerations, and chronic course
MDA5/CADM-140	Classic DM: 10%–15% In adult Japanese: • classic: 10%–45% • clinically amyopathic: 70%–85% • Juvenile DM: 5%–10%	Melanoma differentiation-associated protein 5 (MDA5)/clinically amyopathic DM-140 kDa molecule/IFN-induced with helicase C domain protein 1 (IFIH1)	• Clinically amyopathic DM • Rapidly progressive interstitial lung disease • Cutaneous ulcerations with associated vasculopathy, tender palmar papules, oral pain and ulceration • In juvenile DM, milder muscle disease, mucocutaneous ulcerations, and interstitial lung disease
ARS	Jo-1: 15%–30% PL-7: ≤5% PL-12: ≤5% OJ, EJ, KS, ZO, YRS: rare	Aminoacyl tRNA synthetases (ARS), e.g. Jo-1 targets histidyl tRNA synthetase	• Antisynthetase syndrome with fever, myositis, polyarthritis, Raynaud phenomenon, interstitial lung disease
NXP-2 (p140)	Adult DM: 5%–25% (varies by ethnicity, most common in White population) Juvenile DM: 20%–25%	Nuclear matrix protein 2 (MORC family CW-type zinc finger 3 [MORC3])	• In adult DM, associated with malignancy, subcutaneous edema, and calcinosis • In juvenile DM, more severe muscle disease and calcinosis
SAE	1%–8%	SUMO activation enzyme – posttranslational modification of proteins that are critical for subcellular localization	• Severe cutaneous disease, progressive muscle disease, dysphagia
SRP	1%–8%	Signal recognition particle – intracytoplasmic protein translocation	• Fulminant DM/PM • Cardiac involvement
Low specificity for DM/PM			
ANA (most common IF patterns: speckled, nucleolar)	40%		• Clinically amyopathic DM (65%)
ssDNA	40%	Single-stranded DNA	• SLE, SSc, morphea
PM-Scl (PM-1)	10%	Ribosomal RNA processing enzyme	• Overlap with SSc
SSA/Ro (especially Ro52)	15%	hYRNP	• Overlap with SjS, SCLE, neonatal LE/CHB, SLE
U1RNP	10%	Spliceosome RNP	• Overlap with other AI-CTDs
Ku	3%	DNA end-binding repair protein complex	• Overlap with SSc
U2RNP	1%	Spliceosome RNP	• Overlap with SSc
*Using current assay techniques.			

Table 40.2 Autoantibodies encountered in the idiopathic inflammatory dermatomyopathies (DM/PM). Rarely, patients develop specific aAb targeting Fer (elongation factor 1-α) and Mas (small RNA). ANA, anti-nuclear antibodies; ARS, aminoacyl tRNA synthetase; CADM, clinically amyopathic DM; CHB, congenital heart block; DM/PM, dermatomyositis/polymyositis; IF, immunofluorescence; IFN, interferon; LE, lupus erythematosus; RNP, ribonucleoprotein; SCLE, subacute cutaneous lupus erythematosus; SjS, Sjögren syndrome; SLE, systemic lupus erythematosus; SSc, systemic sclerosis. Adapted from references 2, 32, 36, 54–57.

have little disease specificity and are of little value in the management of patients. The centromeric pattern reflects the binding of aAb to the polypeptide components of chromosomal centromeres (e.g. CENP-B) and is associated primarily with the limited form of SSc (see Table 40.3). A peripheral pattern can be suggestive of aAb to DNA, including dsDNA. The nucleolar pattern is due to aAb binding to ribosomal RNA processing molecules such as fibrillarin and can be indicative of SSc. A speckled pattern is due to various antibodies, including those that recognize Sm, RNP, SSA/Ro, SSB/La, Sc1-70, and RNA polymerases (RNAP). Nowadays, commercial laboratories often report primarily homogeneous and speckled staining patterns because they commonly occur in healthy and older individuals. As a result, a higher threshold (e.g. ≥1 : 160) may need to be considered for determining whether a titer is clinically relevant. In general, ANA patterns have been replaced by specific tests for nuclear and cytoplasmic antigens.

Clinical Utility of the ANA Assay

Key concepts in laboratory testing – sensitivity, specificity, positive predictive value, and negative predictive value – are presented in Fig. 40.4[9]. These concepts should be kept in mind when considering the cost–benefit ratio of laboratory testing for AI-CTD. The ANA test is used most often for the diagnosis of SLE. In SLE patients, its sensitivity and specificity are 98% and 90%, respectively[10], but in unselected patients, the positive predictive value for SLE detection is 30%–40%, while the negative predictive value is 99%. Thus, about

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AUTOANTIBODIES ASSOCIATED WITH SYSTEMIC SCLEROSIS (SSc)				
Target	**SSc, all***	**SSc with diffuse cutaneous scleroderma***	**SSc with limited cutaneous scleroderma (CREST syndrome)***	**Clinical features**
High specificity for SSc				
Centromere (CENP-B)	20%–40%	30%	80%	Better prognosis; pulmonary arterial hypertension
DNA topoisomerase I (formerly Scl-70; DNA-unwinding enzyme)	15%–40%	60%	15%	Worse prognosis; pulmonary fibrosis
RNA polymerase III	5%–30%	45%	6%	Worse prognosis; aAb levels correlate with extent of skin involvement and renal disease; malignancy association, especially breast cancer
Th/To RNP (mitochondrial enzyme)	1%–15%	10%	20%	Worse prognosis; limited pulmonary fibrosis
Fibrillarin (U3RNP, spliceosome RNP)	5%–10%			Worse prognosis; cardiac, renal involvement
RNPC-3 (U11/U12RNP, spliceosome RNP)	3%			Worse prognosis; pulmonary fibrosis, GI involvement, Raynaud phenomenon, malignancy association
Low specificity for SSc				
ANA	95%*			
Angiotensin type 1/Endothelin-1 type A receptors	80%			
PDGFR (expressed on fibroblasts)†	30%–100%			
Endothelial cells (activation and apoptosis)	45%–85%			
MMP-1 and 3 (degrade ECM proteins; aAb prevents degradation)	50%	70%	33%	
HMG (non-histone nucleosomal protein)		30%	40%	
Fibrillin-1 (component of ECM)	>50%			
Histones	40%			
ssDNA	10%–30%			
Rheumatoid factor	25%			
PM-Scl	5%–10%		More commonly associated with limited SSc	Better prognosis; overlap with DM/PM
U1RNP (spliceosome RNP)	2%–15%		More commonly associated with limited SSc	Better prognosis; overlap with MCTD

*Median prevalence using current assay techniques.
†Results vary, depending upon study; evidence that aAb do not activate the receptor; seen in up to 20% of controls.

Table 40.3 Autoantibodies associated with systemic sclerosis (SSc). The hallmark aAb of SSc are shaded in orange. ANA, anti-nuclear antibodies; ECM, extracellular matrix; HMG, high mobility group; IF, immunofluorescence; MCTD, mixed connective tissue disease; PDGFR, platelet-derived growth factor receptor; RNP, ribonucleoprotein; RNPC-3, RNA binding region (RNP1, RRM) containing 3. Adapted from references 2, 49, 50, and 58.

two-thirds of the positive ANA levels observed in a general medical population will be associated with disorders other than SLE, and only very rarely will a person with a negative ANA have SLE. Positive ANA tests are also seen in a large percentage of patients with other AI-CTDs: SSc – 90%; SjS – 70%; and idiopathic inflammatory dermatomyopathies – 40%–65%. There are numerous clinical associations for a positive ANA test[11], and it is noteworthy that clinically silent systemic autoimmune diseases (e.g. autoimmune thyroiditis, pernicious anemia, Addison disease) can contribute to a positive ANA in patients with autoimmune dermatologic disorders such as vitiligo and alopecia areata.

Clinical Value of aAb in Diagnosis, Prognosis, and Disease Activity Monitoring

In everyday practice, these laboratory tests can be used to: (1) establish the diagnosis of an AI-CTD; (2) provide prognostic information; and (3) in specific instances, monitor the level of activity of an AI-CTD[10]. For example, the Sm aAb is highly specific for SLE and thus has considerable diagnostic value. However, the absolute amounts of anti-Sm aAb present in a patient at different time points do not correlate strongly enough with SLE disease activity to have practical clinical value in patient management. Likewise, there is not much value in serial determinations of ANA titers when monitoring disease activity in SLE. However, a few aAb can be used for establishing the diagnosis and monitoring systemic activity, e.g. anti-dsDNA in LE nephritis and cytoplasmic anti-neutrophil cytoplasmic antibodies (c-ANCA) in granulomatosis with polyangiitis. Some aAb can even provide prognostic information, e.g. anti-RNAP aAb are a risk factor for developing renal crisis in patients with systemic sclerosis.

"Sontheimer's Corollary" to Greenwald's Law of Lupus

Robert A. Greenwald formulated a law stating that if a patient is diagnosed as having SLE, everything that happens to that patient afterward, rightly or wrongly, is attributed to SLE. Richard D. Sontheimer formulated a corollary to Greenwald's law, stating that anything happening to a patient with a positive ANA that is not immediately otherwise explicable will automatically be blamed on SLE[12]. This is especially true in patients presenting with non-lupus photosensitive facial eruptions (e.g. rosacea, polymorphic light eruption, photoallergic contact dermatitis) who might also have a positive ANA test for reasons other than SLE[13].

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AUTOANTIBODIES ASSOCIATED WITH OTHER RHEUMATIC DISEASES HAVING CUTANEOUS MANIFESTATIONS			
Autoantibody	**Median prevalence***	**Molecular specificity**	**Clinical associations**
Rheumatoid arthritis			
Rheumatoid factor	80%	Fc portion of IgG	*Low levels*: very nonspecific (healthy elderly individuals, SLE, SSc, MCTD, neoplasia, chronic infections, chronic inflammatory lung diseases, chronic liver disease, sarcoidosis, systemic vasculitides) *High levels*: associated with erosive, crippling RA (25%–90%) and extra-articular manifestations of RA, e.g. systemic vasculitis (i.e. peripheral neuropathy, cutaneous vasculitis); mixed cryoglobulinemia secondary to hepatitis C infection (types II, III; 40%–100%); SjS
Anti-cyclic citrullinated protein (CCP) antibodies	70%	Citrullinated proteins in the skin (e.g., filaggrin) and joints	Increased severity of RA, including radiologic progression of erosive arthritic changes; predictor for the development of RA
Sjögren syndrome			
Alpha-fodrin	70%	An actin-binding protein found at the periphery of chromaffin cells that may be involved in secretion	More specific for SjS than anti-SSA/Ro aAb; also seen in SLE
SSA/Ro	60%	hYRNP	SLE, SCLE, neonatal LE/CHB
SSB/La	20%	hYRNP	SLE, SCLE, neonatal LE/CHB
Mixed connective tissue disease (MCTD, Sharp syndrome)			
U1RNP†	100%‡	Spliceosome RNP	Raynaud phenomenon, swollen hands with sausage-like swelling of digits, polyarthritis, myositis, inverse association with nephritis

*Using current assay techniques.
†In MCTD, associated with high-titer ANA and lack of other aAb.
‡By definition at time of diagnosis.

Table 40.4 Autoantibodies associated with other rheumatic diseases having cutaneous manifestations. More in-depth information concerning clinico-serologic associations can be found elsewhere[8,10,16,59]. aAb, autoantibodies; CHB, congenital heart block; LE, lupus erythematosus; MCTD, mixed connective tissue disease; RA, rheumatoid arthritis; RNP, ribonucleoprotein; SCLE, subacute cutaneous lupus erythematosus; SjS, Sjögren syndrome; SLE, systemic lupus erythematosus; SSc, systemic sclerosis.

Phototherapy and ANA

The question of ANA and anti-SSA/Ro antibody testing often arises in patients being considered for phototherapy, especially if there is any concern about an underlying photosensitive state. However, when screening patients for a multisystem autoimmune disease such as SLE, it is a good rule to remember that patients having any degree of clinically significant SLE are virtually never asymptomatic. Symptoms such as easy fatigability, malaise, and arthralgias are almost always present. In addition, studies have shown that phototherapy protocols such as PUVA or NB-UVB do not significantly induce the appearance of ANA and other aAb[14]. Consequently, while a thorough history, review of systems, and physical examination are indicated in all patients prior to phototherapy, ANA testing is not, unless there is a consistent history or suggestive physical findings.

How Does One Evaluate a Positive ANA Result in a Patient Suspected of Having SLE?

This question would likely be answered differently depending upon the specialist. Dermatologists usually see a highly selected population of LE patients by virtue of their having some form of cutaneous involvement that at times can be the only significant clinical manifestation of the autoimmune disease process (e.g. chronic or subacute cutaneous LE). Therefore, compared to rheumatologists evaluating a patient with suspected SLE, a more focused approach can be undertaken by dermatologists evaluating a positive ANA in a patient who has or is suspected of having cutaneous LE (Fig. 40.5).

AUTOANTIBODIES IN LUPUS ERYTHEMATOSUS

Over 180 LE-associated aAb have been described[15]. Table 40.1 summarizes aAb that are highly specific for SLE as well as less specific aAb that occur in both SLE and other AI-CTD[16]. For example, aAb to dsDNA and Sm are quite specific and the presence of either one represents six points in the EULAR/ACR classification criteria for SLE (see Table 41.5)[17]. The 2012 Systemic Lupus International Collaborating Clinics (SLICC) classification also includes both of these aAb in their immunologic criteria for SLE (see Table 41.4)[18]. Both the EULAR/ACR and SLICC classifications include antiphospholipid antibodies as a criterion, but these are not specific for SLE.

Anti-dsDNA Antibodies

Rising levels of anti-dsDNA aAb correlate positively with SLE disease activity, especially when associated with falling complement values. High levels can reflect an increased risk for LE nephritis. Unfortunately, the clinical value of anti-dsDNA aAb measurements is somewhat diluted by the characteristics of modern assay techniques. The popular ELISA assays preferentially detect *low-avidity* anti-dsDNA aAb that are also seen in patients who do not have SLE. It is recommended that all positive anti-dsDNA aAb ELISA results be confirmed by a second assay technique that detects *high-avidity* anti-dsDNA aAb, such as the Farr radioimmunoassay or the *Crithidia luciliae* indirect immunofluorescence test (CLIFT)[19,20].

Anti-SSA/Ro and Anti-SSB/La Antibodies

Anti-SSA/Ro and anti-SSB/La aAb were originally thought to react exclusively with cytoplasmic ribonucleoproteins (RNP) and were thus designated as cytoplasmic aAb. However, subsequent studies demonstrated that these aAb react with proteins present in human cytoplasmic RNP (hYRNP) that are found in both the cytoplasm and nucleus. Further investigations detected 52 kDa and 60 kDa Ro antigens (termed Ro52 and Ro60); Ro52 is localized within the cytoplasm, while Ro60 is in the nucleus. $Ro52^{-}Ro60^{+}$ serology has been associated with SLE, $Ro52^{+}Ro60^{+}$ with primary SjS, and $Ro52^{+}Ro60^{-}$ with several autoimmune diseases including inflammatory myositis[21]. Because the Ro and La antigens are present on the same RNP particles, an autoimmune response that develops against one of these autoantigens often spreads to the other. This explains the observation that when one aAb is present in an individual, the other is frequently present as well. However, while anti-SSA/Ro aAb can occur alone, this is extremely rare for anti-SSB/La aAb.

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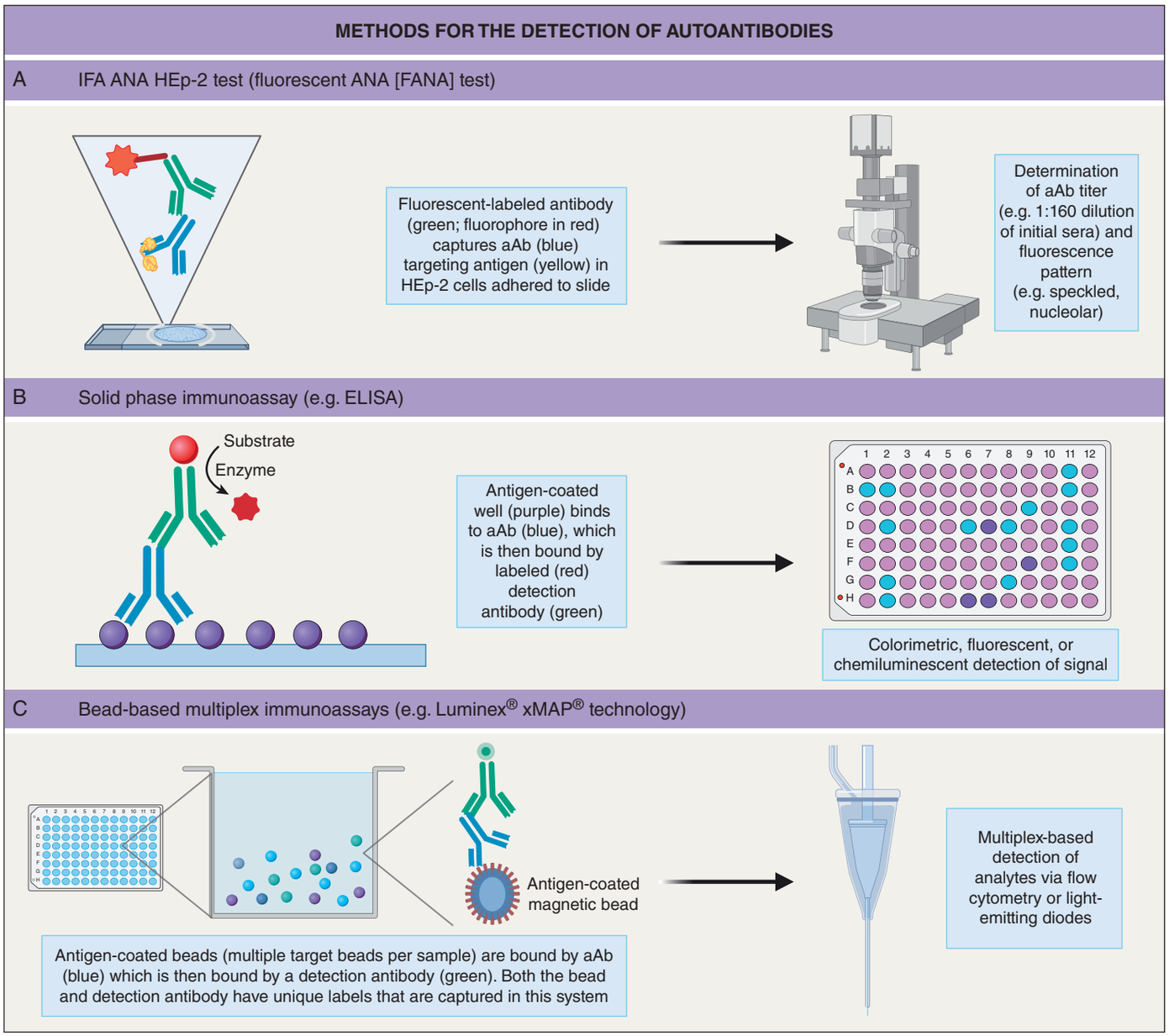

Fig. 40.2 Methods for the detection of ANA. A In the fluorescent ANA (FANA) test, serum samples are usually incubated on a slide with adherent HEp-2 cells. ANA that bind to autoantigens in HEp-2 cells are detected by fluorophore-labeled anti-immunoglobulin (detection) antibodies and visualized with a fluorescence microscope. **B** In a solid-phase immunoassay, a surface (e.g. polystyrene, nitrocellulose) is coated with autoantigens. Following binding of aAb to the autoantigens, they are detected by anti-immunoglobulin (detection) antibodies. By linking the detection antibody to an enzyme that produces one of the following – a colorimetric signal (enzyme-linked immunosorbent assay [ELISA]), a fluorescent signal (fluorometric enzyme-linked immunoassay [FEIA]), or a chemiluminescent signal (chemiluminescence immunoassay [CIA]) – the concentration of aAb is determined. **C** In bead-based multiplex immunoassays, autoantigens are coupled to beads. After the beads are incubated with the patient's serum, bound aAb are detected by fluorophore-labeled detection antibodies in a flow cytometry-based system using two lasers or light-emitting diodes. Both the beads and the detection antibodies have unique fluorescence signals, allowing simultaneous detection of multiple aAb.

MAJOR ISSUES RELATED TO THE INTERPRETATION OF RESULTS OF ANTI-NUCLEAR ANTIBODY (ANA) ASSAYS
• Methods of detecting and reporting ANA (e.g. use of a human tumor cell line substrate leads to increased sensitivity but decreased specificity)
• Subjective nature of FANA assay endpoint
• Requirement for predefined antigen combinations in ELISA or multiplex bead-based assays
• "Normal" versus "abnormal" ANA levels
• Patient age
• Drug-induced ANA
• ANA can be seen in disorders other than AI-CTDs
• Sontheimer's corollary to Greenwald's law of lupus

Table 40.5 Major issues related to the interpretation of results of anti-nuclear antibody (ANA) assays. AI-CTD, autoimmune connective tissue disease; ELISA, enzyme-linked immunosorbent assay.

As with anti-dsDNA aAb assays, it is important to understand certain technical considerations relating to anti-RNP aAb assays, such as those used to measure anti-SSA/Ro, anti-SSB/La, anti-U1RNP, and anti-Sm aAb (i.e. ENA). Nowadays, anti-RNP aAb are usually detected via solid-phase immunoassays such as ELISA (see Fig. 40.2B), which are more sensitive but less specific than the older double immunodiffusion (DID) assays. As a result, some of the original clinical correlations for anti-SSA/Ro, anti-SSB/La, anti-Sm, and anti-U1RNP aAb that were based upon the results of DID assays do not hold with regard to the results of modern ELISA. For example, Lee and colleagues[22] compared the sensitivity of DID assays and ELISA for the detection of aAb to SSA/Ro in patients with SCLE and discoid LE (DLE). High-titer anti-SSA/Ro aAb in the ELISA correlated with positive results in DID assays and the clinical diagnosis of SCLE. Low-titer anti-SSA/Ro aAb in the ELISA were observed in approximately two-thirds of DLE sera tested and correlated with negative results in DID assays. In conclusion, low

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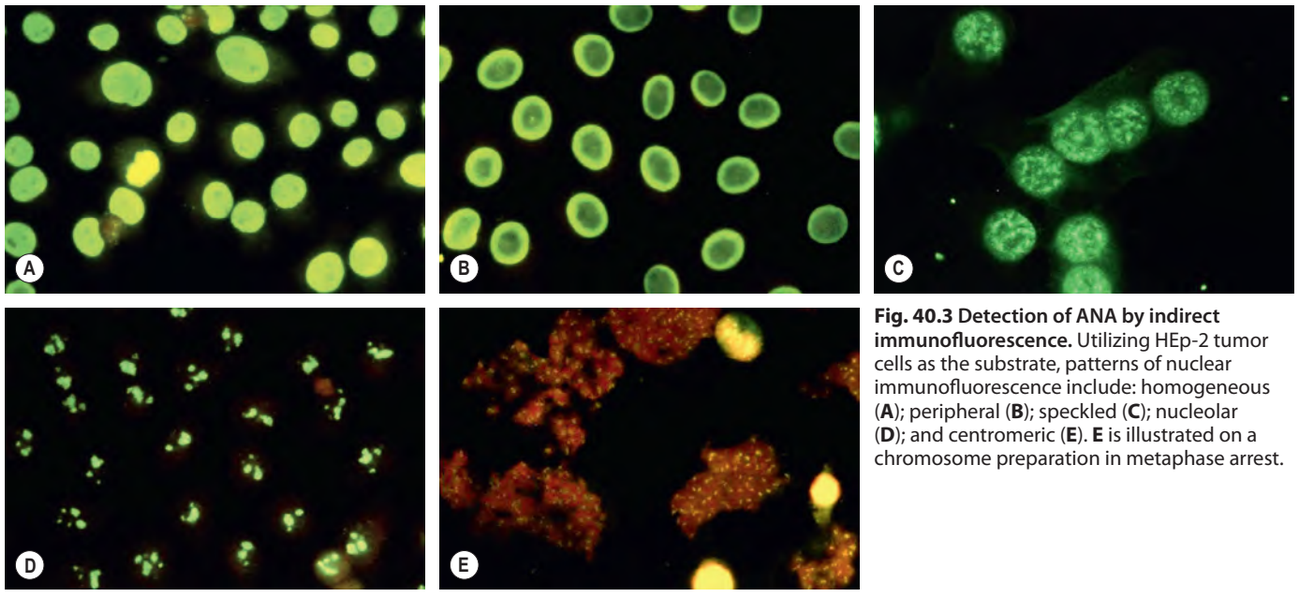

Fig. 40.3 Detection of ANA by indirect immunofluorescence. Utilizing HEp-2 tumor cells as the substrate, patterns of nuclear immunofluorescence include: homogeneous (**A**); peripheral (**B**); speckled (**C**); nucleolar (**D**); and centromeric (**E**). **E** is illustrated on a chromosome preparation in metaphase arrest.

CONCEPT OF CLINICAL UTILITY IN LABORATORY TESTING

	Disease present	Disease absent
Positive result	True positive (TP)	False positive (FP)
Negative result	False negative (FN)	True negative (TN)

Sensitivity=TP/(TP+FN) Positive predictive value=TP/(TP+FP)
Specificity=TN/(TN+FP) Negative predictive value=TN/(TN+FN)

Fig. 40.4 Concept of clinical utility in laboratory testing. *With permission from Ward MM. Laboratory testing for systemic rheumatic diseases. Postgrad Med 1998;103:93–100.*

anti-SSA/Ro titers are of uncertain clinical significance, may be found in normal sera, and should not be used as evidence to support diagnoses classically associated with anti-SSA/Ro aAb.

Drug-Induced ANA/SLE

Several drugs can trigger ANA or a drug-induced SLE syndrome, including minocycline, procainamide, hydralazine, isoniazid, chlorpromazine, phenytoin, quinidine, and methyldopa. The predominant clinical manifestations of drug-induced SLE are musculoskeletal symptoms (arthritis, arthralgia, myalgia) and serositis (pleuritis, pericarditis). Skin involvement is uncommon. In a comprehensive review[23], only 25% of hydralazine-induced SLE patients had "skin involvement", and fewer than 5% of procainamide-induced SLE patients experienced "skin changes", whereas 71% of idiopathic SLE patients had some type of skin change. Anti-histone aAb represent also a serologic marker of drug-induced ANA and drug-induced SLE (present in 95%), but can also be present in patients with idiopathic SLE.

A different set of drugs, including hydrochlorothiazide, terbinafine, proton pump inhibitors, griseofulvin, diltiazem and taxanes, can trigger anti-SSA/Ro aAb production and the cutaneous lesions of SCLE[24]. Anti-histone aAb are not a characteristic finding in drug-induced SCLE. Finally, SCLE, DLE, and SLE have occasionally been observed in patients receiving immune checkpoint inhibitors such as nivolumab and pembrolizumab[25].

Induction of ANA and anti-dsDNA antibodies by tumor necrosis factor inhibitors

Tumor necrosis factor (TNF) blockade by recombinant proteins is a common therapeutic intervention (see Ch. 128) and is associated with the induction of aAb, especially ANA and anti-dsDNA antibodies. However, the clinical significance of these aAb is not well defined and the incidence of clinically detectable autoimmune disease is low. In a study of 125 psoriasis patients treated with a TNF inhibitor, newly positive ANA titers developed after a mean of 8–20 months in ~20% of those who received etanercept (9/45) or adalimumab (9/47), compared with 48% of those who received infliximab (16/33); anti-dsDNA aAb developed in 6% of patients in each group (8/125 overall)[26]. However, clinical manifestations of LE were not evident in any of these patients. In a series of 22 patients who developed a lupus-like syndrome during therapy with either infliximab or etanercept, cutaneous features included: a butterfly rash (ACLE), SCLE, DLE, small vessel vasculitis, and chilblains[27]. Additional findings were arthritis, phlebitis (with anti-cardiolipin antibodies), leukopenia, thrombocytopenia, serositis, and general manifestations (e.g. fever, weight loss, asthenia); renal involvement was rare[28].

Notably, all the TNF inhibitors have the propensity to induce ANA and anti-dsDNA aAb as well as cutaneous LE, SLE, or a lupus-like syndrome. When a French pharmacovigilance database was analyzed for TNF inhibitor-induced LE or lupus-like syndrome, the estimated reporting odds ratios were 9–10 for infliximab and adalimumab, and 4 for etanercept[29]. Interestingly, in patients with rheumatoid arthritis and psoriasis, the production of lupus-associated antibodies was linked to both the formation of anti-drug antibodies (resulting in decreased therapeutic efficacy) and type 1 interferon production[30]. However, when over 1000 patients with rheumatoid arthritis underwent serial measurements of ANA and anti-dsDNA aAb, the serologic results failed to predict the onset of lupus or vasculitis[31].

In summary, although the development of aAb is relatively common during TNF inhibitor administration, they are only occasionally of clinical significance. In the absence of clinical signs or symptoms, we do not recommend screening for ANA prior to initiation of TNF inhibitors.

AUTOANTIBODIES IN IDIOPATHIC INFLAMMATORY DERMATOMYOPATHIES

The term "idiopathic inflammatory *dermato*myopathies" (syn. dermatomyositis [DM]/polymyositis [PM], idiopathic inflammatory myopathies; see Ch. 42) has been coined to include cutaneous-only or cutaneous-predominant subsets of DM, such as clinically amyopathic DM (i.e. amyopathic DM and hypomyopathic DM). Patients with adult-onset, clinically amyopathic DM (CADM) appear to have a somewhat higher prevalence of ANA (65%) than patients with classic DM and PM. Compared to LE, there are more highly specific aAb, and titers of some of these specific aAb are associated with disease severity[32]. Table 40.2

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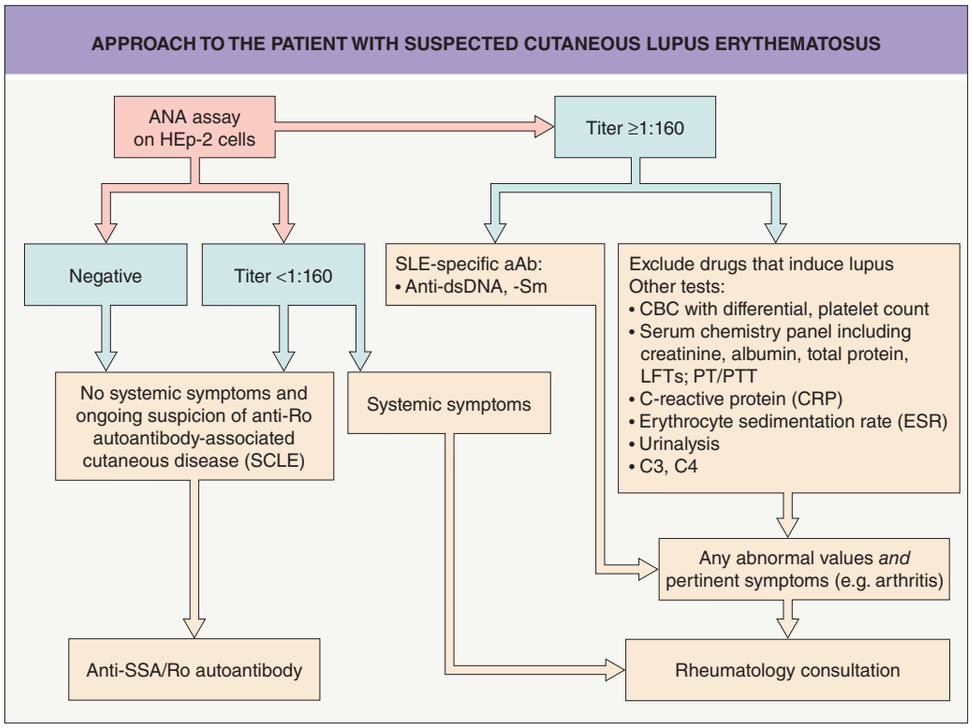

Fig. 40.5 Approach to the patient with suspected cutaneous lupus erythematosus. It should be noted that these recommendations are most relevant within the context of a dermatology practice setting. CBC, complete blood count; LFTs, liver function tests; PT, prothrombin time; PTT, partial thromboplastin time; SCLE, subacute cutaneous lupus erythematosus.

summarizes the aAb in patients with idiopathic inflammatory dermatomyopathies. Both high specificity and low specificity aAb are included, with the latter also seen in other AI-CTD.

Anti-TIF1-γ (p155), -TIF1-α (p140) Antibodies

aAb to a 155 kDa polypeptide was initially reported in 20%–30% of patients with the classic form of juvenile-onset or adult-onset DM[33] and was subsequently noted to bind to transcriptional intermediary factor 1 gamma (TIF1-γ). aAb to TIF1-α can also be observed in DM and are associated with TIF1-γ antibodies (see Table 40.2). In patients with adult-onset classic DM, the presence of anti-TIF1-γ aAb has been associated with an increased risk of internal malignancy[34]. Individuals with anti-TIF1-γ aAb often have diffuse erythema in a photodistribution, extensive inflammatory skin disease, and relatively mild muscle disease; calcinosis is uncommon[32]. They also have palatal erythema, a finding termed "ovoid patch"[35], in addition to keratotic palmar papules, psoriasiform lesions, and hypopigmented and telangiectatic ("red on white") patches[36].

Anti-MDA5/CADM-140 Antibodies

In adult Japanese patients with clinically amyopathic DM (CADM) and interstitial lung disease, aAb to a 140 kDa molecule (CADM-140) were first noted[37]. Subsequently these aAb were found to bind the interferon-induced with helicase C domain protein 1 (IFIH1), which is also known as melanoma differentiation-associated protein 5 (MDA5)[38]. In a retrospective series, plasma samples from 77 patients with DM were screened for anti-MDA5 aAb[39]. The ten patients (13%) with this aAb had a rather distinct clinical phenotype with cutaneous ulcerations that can overlie Gottron papules, tender palmar papules, oral pain and ulceration, arthritis/arthralgia, and diffuse hair loss. Recognition of this subset of patients is clinically important because of the association of anti-MDA5 aAb with rapidly progressive interstitial lung disease; the latter occurs more frequently in Asians[32]. The titer of anti-MDA5 aAb is used to monitor the associated interstitial lung disease and the risk of its relapse following therapy[40].

Anti-Jo-1 Antibodies

The anti-Jo-1 antibody is one of several myositis-specific aAb directed against the aminoacyl transfer RNA synthetases (ARS) that deliver amino acids to ribosomes during protein translation. The presence of one or more of these anti-ARS aAb correlates positively with a subset of mostly PM and some DM patients who have an increased risk of erosive arthritis, Raynaud phenomenon, and interstitial lung disease; this clinico-serologic constellation has been referred to as the "antisynthetase syndrome"[41]. The mechanic's hand skin lesion was originally described in these patients, but has subsequently been observed in other DM subsets, including CADM[42].

Anti-Mi-2, -SRP, -NXP-2, and –SAE Antibodies

Anti-Mi-2 aAb are found in patients with the hallmark skin lesions of DM (Gottron papules, Gottron sign, shawl sign). aAb to the signal recognition particle (SRP) are uncommon, but when present, they signal a risk for fulminant DM/PM with cardiac involvement and a high mortality rate. Anti-nuclear matrix protein 2 (NXP-2) aAb are associated with more severe muscle disease and calcinosis in children[32]; in adults, they are associated with subcutaneous edema, increased risk of cancer, and calcinosis[43,44]. aAb to SAE (SUMO activation enzyme) have been described in a rare subset of DM patients with severe cutaneous disease, progressive muscle involvement, and dysphagia[32].

AUTOANTIBODIES IN SYSTEMIC SCLEROSIS (SSc; SCLERODERMA) AND MORPHEA (LOCALIZED SCLERODERMA)

Table 40.3 summarizes aAb associated with SSc (see Ch. 43)[1,45,46]. The current classification criteria for SSc incorporate the following aAb: anti-centromere (CENP-B); anti-topoisomerase I (formerly Scl-70); and anti-RNA polymerases (RNAP), in particular anti-RNA polymerase III[47]. Anti-centromere aAb correlate with the limited cutaneous form of SSc (CREST syndrome), while aAb against DNA topoisomerase I and RNAP are seen more commonly in the diffuse cutaneous form of SSc. aAb that recognize RNAP are also associated with an increased risk of developing SSc-associated renal disease[48]. Anti-RNA polymerase III aAb have been associated with an increased risk of internal malignancies and may be indicative of paraneoplastic SSc[49,50], with triggering of cellular immunity by the malignancy also promoting sclerosis[50]. However, these aAb are not subset- or disease-specific; e.g. anti-CENP-B aAb can also be detected in ~10% of patients with diffuse cutaneous SSc as well as in disorders such as primary biliary cholangitis. Lastly, as with other AI-CTDs, some aAb encountered in SSc patients are nonspecific (e.g. anti-histone, anti-ssDNA, anti-PM/Scl).

While there is little clinical overlap between SSc and morphea (see Ch. 44), some degree of serologic overlap has been described (e.g.

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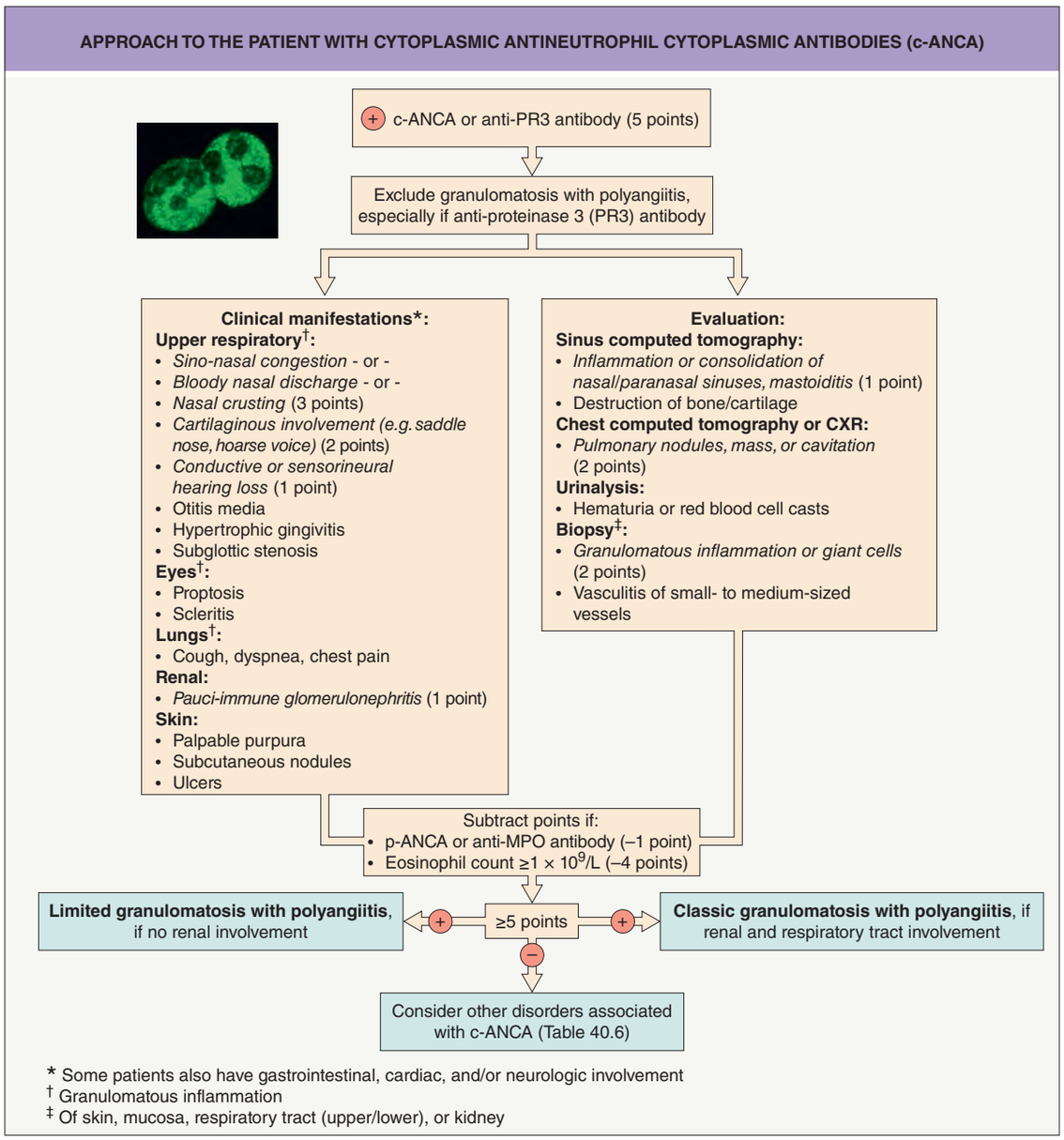

Fig. 40.6 Approach to the patient with cytoplasmic anti-neutrophil cytoplasmic antibodies (c-ANCA). The presence of ≥5 points yields a sensitivity of 93% and specificity of 94% for GPA per the ACR/EULAR 2022 classification criteria. *Photomicrograph from Wiik AS, Fritzler MJ. Laboratory tests in rheumatic disorders. In: Hochberg MC, Silman AJ, Smolen JS, et al. (eds). Rheumatology, 4th edn. London: Mosby Elsevier; 2008.*

anti-fibrillin-1 aAb). aAb to fibrillin-1 (as distinguished from fibrillarin [U3RNP]), histones, and ssDNA occur in 30%–50% of patients with morphea. As with classic ANA, these antibodies have been observed more often in patients with more widespread forms of morphea such as linear or generalized morphea. To date, morphea-specific aAb have not been identified nor have aAb that reflect disease activity or prognosis.

AUTOANTIBODIES IN OTHER RHEUMATIC DISORDERS AFFECTING THE SKIN

Table 40.4 summarizes aAb seen in patients with other rheumatic disorders that can have cutaneous manifestations (see Ch. 45)[1]. Anti-neutrophil cytoplasmic antibodies (ANCA) are a subgroup of neutrophil-specific aAb that are characteristic of systemic vasculitides (see Ch. 24). Figs. 40.6 and 40.7 present a clinical approach to patients with either cytoplasmic ANCA (c-ANCA) or perinuclear ANCA (p-ANCA) and Table 40.6 summarizes other disorders associated with ANCA.

Acknowledgment

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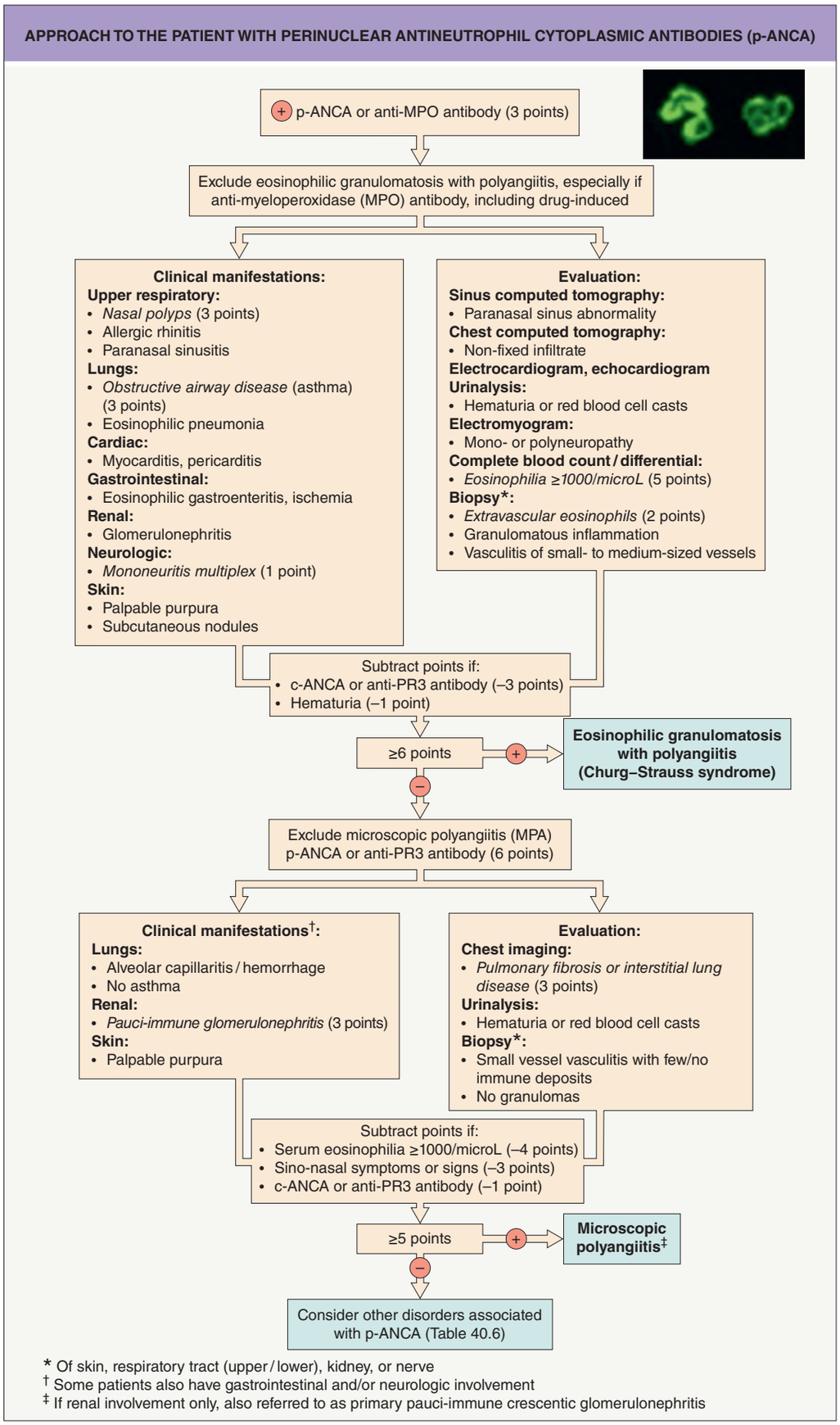

Fig. 40.7 Approach to the patient with perinuclear anti-neutrophil cytoplasmic antibodies (p-ANCA). Use of leukotriene inhibitors (see Ch. 130) has been associated with the onset of eosinophilic granulomatosis with polyangiitis (EGPA; Churg–Strauss syndrome). Whether this represents unmasking of previously undiagnosed EGPA is a subject of ongoing debate. The presence of ≥6 points yields a sensitivity of 85% and a specificity of 99% for EGPA while >5 points yields a sensitivity of 91% and a specificity of 94% for microscopic polyangiitis per the ACR/EULAR 2022 classification criteria. *Photomicrograph from Wiik AS, Fritzler MJ. Laboratory tests in rheumatic disorders. In: Hochberg MC, Silman AJ, Smolen JS, et al. (eds). Rheumatology, 4th edn. London: Mosby Elsevier; 2008.*

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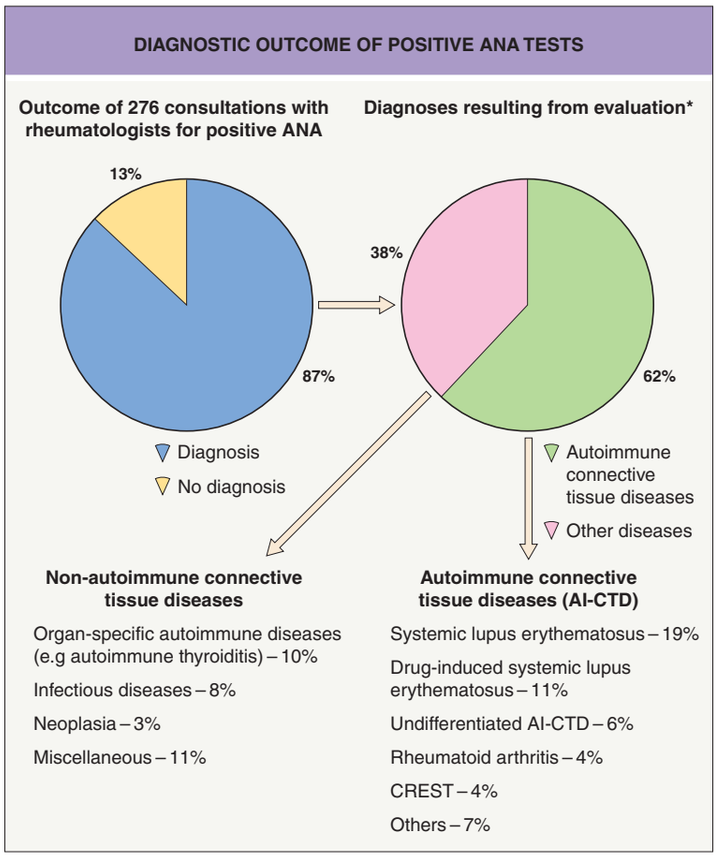
OTHER DISORDERS ASSOCIATED WITH ANTI-NEUTROPHIL CYTOPLASMIC ANTIBODIES (ANCA)			
	Type of ANCA*	**Major antigen(s)†**	**Evaluation**
Inflammatory bowel disease			
Ulcerative colitis (50%–70%) > Crohn disease (20%–40%)	p > c	Catalase, BPI, α-enolase, LF, CG, HMG1/2 > MPO, PR3	Colonoscopy
Autoimmune liver disease			
Autoimmune hepatitis (type 1; 80%), primary sclerosing cholangitis (70%) > primary biliary cirrhosis (30%)	p > c	Actin, HMG1/2	LFTs; ANA; anti-smooth muscle Ab; antimitochondrial Ab
Autoimmune connective tissue disease			
Systemic lupus erythematosus, rheumatoid arthritis, systemic sclerosis (20%–30%) > Sjögren syndrome, dermatomyositis, reactive arthritis, relapsing polychondritis, antiphospholipid syndrome	p > c	MPO, PR3, BPI, CG, LF, lysozyme, HMGI/2	ANA panel; RF; anti-cardiolipin and anti-β_2 glycoprotein I Ab
Drug-induced vasculitis/lupus/hepatitis			
Propylthiouracil (20%), minocycline, methimazole, carbimazole > levamisole, sulfasalazine	p > c	MPO > PR3, elastase	Skin biopsy; ANA panel; LFTs; discontinue drug
Hydralazine, penicillamine > allopurinol‡	p	MPO, elastase	
TNF inhibitors	p, c	MPO, PR3	
High-dose intravenous immunoglobulin	c	–	
Other cutaneous or systemic vasculitides¶			
Mixed cryoglobulinemia, Behçet disease, paraneoplastic	p > c	Variable	Skin biopsy; cryoglobulins; hepatitis B, C panels; exclude malignancy
Henoch–Schönlein purpura, Buerger disease	p	Variable	
Other glomerulonephritides			
Anti-GBM Ab-associated (30%) > immune complex (15%) > poststreptococcal	p	MPO	UA; BUN/creatinine; ASO; renal biopsy
Infections/Vaccines			
Respiratory tract infections (bacterial/mycobacterial/ fungal), enteritis, periodontitis, HIV, parvovirus B19, malaria, leprosy, SARS-CoV-2 infection, and SARS-CoV-2 mRNA vaccine	p, c	MPO, others	Cultures; serologies; echocardiogram
Onchocerciasis	p	Defensin	
Hepatitis C virus infection	p, c	CG, BPI, MPO	Serologies
Subacute bacterial endocarditis, chromomycosis, invasive amebiasis	c	PR3	
Miscellaneous			
Cystic fibrosis (80%)	c > p	BPI	Sweat test
Hypergammaglobulinemia	p, c	–	Serum protein electrophoresis

*c-ANCA are often atypical; in addition, immunofluorescence testing in some laboratories may not distinguish c-ANCA (atypical) from atypical ANCA.
†p-ANCA often have antigen specificity other than MPO, e.g., LF, elastase, CG, lysozyme, BPI, HMG1/2, α-enolase, catalase, actin, defensin; c-ANCA (atypical) have antigen specificity other than proteinase 3, e.g. BPI, lysozyme or multiple antigens.
‡Other drugs associated with p-ANCA in scattered case reports include cimetidine, cefotaxime, phenytoin, isotretinoin, clozapine, atorvastatin, indomethacin, isoniazid, and thioridazine.
¶Although primarily associated with c-ANCA, 5%–50% of patients with granulomatosis with polyangiitis have p-ANCA; likewise, although primarily associated with p-ANCA, 5%–30% of patients with eosinophilic granulomatosis with polyangiitis and ~25%–30% of those with microscopic polyangiitis have c-ANCA; ANCA have also been reported in leukocytoclastic vasculitis of unknown etiology and unclassified vasculitis.

Table 40.6 Other disorders associated with anti-neutrophil cytoplasmic antibodies (ANCA). Ab, antibodies; ANA, anti-nuclear antibodies; ASO, antistreptolysin O; BPI, bactericidal/permeability-increasing protein; c, cytoplasmic [c-ANCA or c-ANCA (atypical)]; CG, cathepsin G; GBM, glomerular basement membrane; HMG1/2, high mobility group non-histone chromosomal proteins; LF, lactoferrin; LFTs, liver function tests; MPO, myeloperoxidase; p, perinuclear (p-ANCA); RF, rheumatoid factor; UA, urinalysis. *Courtesy Julie V. Schaffer, MD.*



eFig. 40.1 Diagnostic outcome of positive ANA tests. Outcome of rheumatology evaluations of patients referred for consultations for positive ANA of undetermined clinical significance. *Total number of patients = 276, but number of diagnoses = 284, because 8 patients had more than one diagnosis. *Adapted from data presented by Tan EM, et al. Arthritis Rheum 1997;40:1601–11*.

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